

Olivia Kaitlyn Roberts, MS – Curriculum Vitae

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EDUCATION

University of Michigan, Ann Arbor, MI Aug 2024 – May 2026
M.S. Computational Epidemiology and Systems Modeling
Certificate in Complex Systems GPA: 3.9

University of South Dakota, Vermillion, SD Aug 2021 – May 2024
B.S. Mathematics
Minors in Music and Interdisciplinary Sciences GPA: 3.9

RESEARCH EXPERIENCE

Epidemiology Research Assistant, University of Michigan School of Public Health May 2025 – Present

- Conducts research with Dr. Andrew Brouwer for the Center for the Assessment of Tobacco Regulations (CAsToR)
- Applies multistate transition models using complex survey weights in R to analyze longitudinal trends in tobacco use among adults and youth in nationally representative studies

UDiscover Summer Research Scholar, University of South Dakota May 2023 – Aug 2023

- Conducted pure mathematics research under Dr. Gabriel Picioroaga on generalizing music theory using algebra
- Developed mathematical proofs and Maple programs to construct new musical systems

Behavioral Health Epidemiology Intern, South Dakota Department of Social Services May 2022 – Aug 2022

- Analyzed state-level data on substance use and mental health using R, Excel, and Power BI
- Contributed to behavioral health data and outcomes reports for state government and public use

TECHNICAL SKILLS

- Programming: R, Python, Matlab, Stata
- Tools: LaTeX, Power BI, Excel, SEER Stat, SQL
- Markov multistate transition modeling
- Network modeling: contact networks, outbreak simulation
- Infectious and chronic disease modeling: deterministic and stochastic models

PUBLICATIONS

- Picioroaga G, **Roberts O**. Musical systems with $\mathbb{Z}_n$ – Cayley graphs. *MusMat Brazilian Journal of Music and Mathematics*. 2024;8(1). doi:10.46926/musmat.2024v8.1-23.
- Brouwer AF, **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R. Transitions in cigarette and ENDS use in the PATH Study: a multistate transition model analysis of adults in 2021–2022 compared to previous years. *Manuscript submitted*
- **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R, Brouwer AF. Transitions in cigarette, ENDS, and dual use among youth in the PATH Study from 2015–2023: a multistate transition modeling analysis. *Manuscript submitted*
- **Roberts OK**, Bover Manderski MT, Gonsalves NJ, Jeon J, Meza R, Delnevo CD, Brouwer AF. Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus Study, 2022–2025. *Manuscript in preparation*

PRESENTATIONS

“Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus study, 2022–2025”

- Poster presentation, University of Michigan Epidemiology Poster Session, Apr 2026
- Poster presentation, NIH Tobacco Regulatory Science Meeting, July 2026

“Transitions in cigarette, ENDS, and dual use among youth in the PATH Study from 2015–2023: a multistate transition modeling analysis”

- Invited seminar talk (1 hour), University of South Dakota Department of Mathematics, Mar 2026
- Oral presentation, Society for Research on Nicotine and Tobacco (SRNT) Meeting, Mar 2026

“Musical Systems with $\mathbb{Z}_n$ – Cayley Graphs”

- Oral presentation, Joint Mathematics Meetings (JMM) Pi Mu Epsilon Session, Jan 2024
- Oral presentation, Nebraska Conference for Undergraduate Women in Mathematics (NCUWM), Jan 2024
- Oral presentation, National Conference on Undergraduate Research (NCUR), Apr 2023 and Apr 2024

HONORS & AWARDS

Research Awards and Grants

- Pi Mu Epsilon Outstanding Student Speaker Award, Joint Mathematics Meetings, Jan 2024
- Undergraduate Research Excellence Award, University of South Dakota, Sept 2023
- UDiscover Summer Research Grant, University of South Dakota, May 2023
- John W. Carlson Research Grant, University of South Dakota, Apr 2023

Scholarships

- Epidemiology Student Scholarship, University of Michigan, Feb 2024 (75% of tuition)
- South Dakota NASA Space Grant Consortium Educational Stipend, Oct 2023
- Dr. Cleland Cook Mathematics Scholarship, University of South Dakota, Apr 2022 and Apr 2023

TEACHING EXPERIENCE

Yoga Teacher, University of Michigan Recreation

Aug 2024 – Present

- Leads restorative and vinyasa yoga classes to 20+ students multiple times a week
- Developed a 6-week Elevate Yoga workshop series to deepen students' yoga practice

Instructor Aide, University of Michigan Department of Mathematics

July 2024 and July 2025

- Co-instructed three summer camp sessions for 15-20 high school students for the Michigan Math and Science Scholars program
- Assisted with curriculum development and delivered lectures on group theory, set theory, and music theory

Instructor Assistant, University of Michigan School of Nursing

Jan 2025 – May 2025

- Supported 100 students in NURS 218 (Applied Statistics)
- Held office hours, graded assignments, assisted students with Stata programming, and led review sessions

Math Tutor, University of South Dakota

Jan 2022 – May 2024

- Tutored undergraduate students in algebra, statistics, and calculus

Yoga Teacher , University of South Dakota Wellness Center	Aug 2021 – May 2024
- Led sculpt and vinyasa yoga classes for 20+ students multiple times per week - Organized the first and second annual USD Yoga Festivals and led multiple yoga workshops	

OTHER

Event Volunteer , VegMichigan	June 2024 – Present
Admissions Ambassador Volunteer , University of Michigan School of Public Health	Aug 2024 – May 2026
200-Hour Yoga Teaching Certification , Tiny Buddha Yoga, Ann Arbor, MI	Dec 2024
CPR/AED for Professional Rescuers Certification	Sept 2024
200-Hour Yoga Teaching Certification , Better Together Yoga, Sioux Falls, SD	Sept 2021